

and we cannot obtain an integral expression ignoring the relation between the angular functions. This difference between the cases occurs because $p\omega$ and p' are parallel quaternions, while pj and p' are not. Only parallel quaternions admit a simple angular polar function.

By way of example the quaternionic curve

$$q(t) = R(e^{i\phi} + e^{i\psi}j) + R[\cos \vartheta e^{i\phi} + \sin \vartheta e^{i\psi}j], \quad \vartheta \in [0, 2n\pi] \quad (25)$$

with smooth and continuous $\phi = \phi(t)$, and $\psi = \psi(t)$, and $n \in \mathbb{Z}$. In the particular case where

$$p_0(t) = R(e^{i\phi} + e^{i\psi}j), \quad \phi = \psi, \quad \text{and} \quad p = q - p_0 \quad (26)$$

$p = q - p_0$, then

$$\langle pj, p' \rangle = R^2 \vartheta'. \quad (27)$$

As in the first case, we have to ascertain the parameters in order to have the correct integration, where $\vartheta \in [0, \pi/2]$. Hence, there are three changes to be done,

$$\begin{aligned} \text{i.} \quad & [\frac{\pi}{2}, \pi] \rightarrow [\frac{\pi}{2}, 0], \quad \text{and} \quad \phi \rightarrow \phi + \pi, \\ \text{ii.} \quad & [\pi, \frac{3\pi}{2}] \rightarrow [0, \frac{\pi}{2}], \quad \phi \rightarrow \phi + \pi \quad \text{and} \quad \psi \rightarrow \psi + \pi, \\ \text{iii.} \quad & [\frac{3\pi}{2}, 2\pi] \rightarrow [\frac{\pi}{2}, 0], \quad \text{and} \quad \phi \rightarrow \phi + \pi. \end{aligned} \quad (28)$$

Consequently, the four integrals to be calculated are

$$\frac{1}{R^2} \int_0^{\pi/2} \langle pj, p' \rangle d\vartheta = \frac{\pi}{2}. \quad (29)$$

Repeating the process $4n$ times, we obtain the total rotation number to be $2n\pi$, as expected. Definition 3.2 of the winding number also holds in the symplectic case, although choices such as $\psi = \phi$ have to be made.

4 WINDING NUMBER AND HOMOTOPY

As in the previous section, we keep the complex curves as the model for the quaternionic curves case. Thus, we define the parameter α of deformation of a quaternionic curve, so that

$$J \subset \mathbb{R}, \quad 0 \in J \quad \text{and} \quad \alpha \in J. \quad (30)$$

In analogy to complex curves:

Definition 4.1 (Deformed curve) Let $q(t) : I \rightarrow \mathbb{H}$ be a continuous curve parametrized as $\mathbb{R} \times \mathbb{R}\omega$. The regular curve $p(t, \alpha) : I \times J \rightarrow \mathbb{H}$ is a continuous deformation of q if it satisfies

$$p(t, 0) = q(t).$$

As it is well known from plane curves, the angular function $\theta(t, \alpha)$ associated to the deformation is also continuous in the case of a fixed value $\alpha_0 \in J$. The concept of continuous deformation permits the introduction of the spatial concept of homotopy, in analogy to the plane concept. In order to get it, let us then prove several necessary results.

Lemma 4.1 Let $q(\alpha, t) : J \times I \rightarrow \mathbb{H}$ be a continuous deformation of the regular curve $q(t) : I \rightarrow \mathbb{H}$, and the regular curve $p(\alpha) : J \rightarrow \mathbb{H}$ is such that $p(\alpha, t) \cap q(\alpha, t) = \emptyset$. The angular function $\theta(\alpha, t)$ associated to the points of $q(\alpha, t)$ using the point $p(\alpha)$, such that $\theta(\alpha, \alpha) = 0$, is a continuous function of t and α

Proof: Let us define the unit vector

$$v_\alpha(t) = \frac{q(\alpha, t) - p(\alpha, t)}{|q(\alpha, t) - p(\alpha, t)|}$$

Let us suppose that, given $r, s \in I$ we have that $\langle v_\alpha(t), v_\alpha(s) \rangle \neq -1$. The continuity of $q(\alpha, t)$ permit us to built the sequences that obbey

$$|\alpha_0 - \alpha| < \frac{1}{n}, \quad \text{and} \quad |t_n - s_n| < \frac{1}{n}, \quad \text{where} \quad \alpha_0 \in J \quad \text{and} \quad n \in \mathbb{N}.$$

consequently,

$$\lim_{n \rightarrow \infty} \alpha_n = \alpha_0 \quad \text{and} \quad \lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} t_n = t_0 \in I$$

Consequently, exists $\delta > 0$ so that $|\alpha_0 - \alpha| < \delta$ and $|s - t| < \delta$ so that $\langle v_\alpha(t), v_\alpha(s) \rangle \neq -1$, a contradiction, and thus we can use the inner product build from the inner product is a continuous function. If we divide I in segments such that $a = t_0 < t_1 < \dots < t_n = b$, the have that $|t_{k+1} - t_k| < \delta$, and $|\alpha_0 - \alpha| < \delta$ define the angular function as a sum of continuous functions over k , and thus $\theta(\alpha, t)$ is continuous. $\square$

The above proof is identical to he complex result, the only difference is the definition of the rotation angle associated to either to Proposition 3.2 or the inner product (24). Thus we can follow the analogy with another result.

Theorem 4.1 *Let $q(\alpha, t) : J \times I \rightarrow \mathbb{H}$ be a continuous deformation of the regular curve $q(t) : I \rightarrow \mathbb{H}$, where $J = [0, \infty)$. If $p(\alpha) : J \times I \rightarrow \mathbb{H}$ is a regular a continuous curve such that $p(\alpha, t) \cap q(\alpha, t) = \emptyset$, the winding number $\mathcal{R}(q, p)$ is a constant independent of α .*

Proof: From Proposition 3.3 and Lemma 4.1, the winding number

$$\mathcal{R}(q(\alpha, b), p(\alpha, b)) = \frac{1}{2\pi} \theta(\alpha, b)$$

is a continuous function of α . However, the winding number assumes only discrete values, and thus it is a constant independent of α . $\square$

The above theorem is also valid in the case of symplectic curves, as well. An immediate consequence of the Proposition 3.3 and Theorem 4.1 is

Corollary 4.1 *Let $q(\alpha, t) : J \times I \rightarrow \mathbb{H}$ be a continuous deformation of the regular curve $q(t) : I \rightarrow \mathbb{H}$, where $J = [0, \infty)$. Let $p(\alpha) : J \rightarrow \mathbb{H}$ be a continuous curve such that $p(\alpha, t) \cap q(\alpha, t) = \emptyset$, and $\lim_{\alpha \rightarrow \infty} |p(\alpha)| = \infty$. Thus*

$$\mathcal{R}(q, p(0)) = 0.$$

The deformation of complex curves can be immediately adopted to quaternionic curves from Definition 4.1, and we can define:

Definition 4.2 (Homotopy) *Let $p, q : I \rightarrow \mathbb{H}$ be two closed quaternionic curves. If the regular application $h(\alpha, t) : [0, 1] \times I \rightarrow \mathbb{H}$ is a closed curve for every value of $\alpha \in J$ such that $h(0, t) = p(t)$ and $h(1, t) = q(t)$, is called an homotopy between p and q , which are then called homotopic.*

Another immediate consequence of Theorem 4.1 is

Corollary 4.2 *Let $p, q : I \rightarrow \mathbb{U}$ be two closed regular homotopic quaternionic curves and $\mathbb{U} \subset \mathbb{H}$. If the regular curve $p_0(t) \notin \mathbb{U}$, thus $\mathcal{R}(p, p_0) = \mathcal{R}(q, p_0)$.*

The winding number and the homotopy are related in the following theorems, that are analogous to plane curves.

Theorem 4.2 (Quaternionic Poincaré-Bohl theorem) *Let $p_0 : I \rightarrow \mathbb{H}$ be a regular quaternionic curve, and the regular closed curves $p, q : I \rightarrow \mathbb{H} \setminus \{p_0(t)\}$. If $p_0(t)$ does not intercept any segment that connects $p(t)$ and $q(t) \forall t \in I$, then, $\mathcal{R}(p, p_0) = \mathcal{R}(q, p_0)$.*

Proof: Building the homotopy $h = \alpha q + (1 - \alpha)p$ and evoking Corollary 4.2, the theorem is proven. $\square$

Immediately, we have,

Corollary 4.3 *Let $p, q : I \rightarrow \mathbb{U}$ be two closed regular homotopic quaternionic curves and $\mathbb{U} \subset \mathbb{H}$. If $|p - q| < |q - p_0|$, and $|p - q| < |p - p_0|$ hold $\forall t \in I$, then $\mathcal{R}(p, p_0) = \mathcal{R}(q, p_0)$.*

Finally, we may obtain two interesting results that are analogous of complex analysis.

Theorem 4.3 *Let $\omega(t), p_0(t) : I \rightarrow \mathbb{H}$ be regular quaternionic functions. For each particular value of $t \in I$, let $D_{\mathbb{H}}$ be a disc of radius R , center p_0 , and border $C_{\mathbb{H}}$ in the $\mathbb{R} \times \mathbb{R}\omega$ plane and let $F : D_{\mathbb{H}} \rightarrow \mathbb{R} \times \mathbb{R}\omega$ be a continuous function. If the winding number is such that $\mathcal{R}(F(S_{\mathbb{H}}), p_0) \neq 0$, there is a point $q_0 \in D_{\mathbb{H}}$ such that $F(q_0) = p_0$.*

Proof: Let $s(\alpha, t) = p_0(t) + \alpha r(\cos(2n\pi t) + \omega \sin(2n\pi t))$ be a circle in $\mathbb{R} \times \mathbb{R}\omega$, where $n \in \mathbb{Z}$ and of course $\mathcal{R}(s, p_0) = n$. Let us suppose that $p_0 \notin F(D_{\mathbb{H}})$. Thus, the function $h(\alpha, t) = F(s(\alpha, t)) - p_0$ is an homotopy between F and the constant $F(p_0)$ such that every curve belongs to a subspace of $\mathbb{H} - \{p_0\}$. Consequently, $F(S_{\mathbb{H}})$ is homotopic to a point, whose winding number is zero and consequently $\mathcal{R}(F(S_{\mathbb{H}}), p_0) = \mathcal{R}(F(p_0), p_0) = 0$. This contradicts the hypothesis, and thus there is a point q_0 such that $F(q_0) = p_0$. In particular, we can choose $p_0 =$ and obtain $F(q_0) = 0$. $\square$

Finally, we have:

Theorem 4.4 (Fundamental theorem of algebra for quaternions) *The quaternionic polynomial*

$$F(q) = \sum_{\ell=0}^n a_{n-\ell}(q - q_0)^\ell = 0, \quad \text{where} \quad a_\ell \in \mathbb{R} \quad \text{and} \quad q_0 \in \mathbb{H},$$

admits a quaternionic root.

Proof: We restrict $F(q)$ to the circumference C_{q_0} of center q_0 of radius $R = 1 + \sum_{\ell=0}^n |a_\ell|$. Using $q - q_0 = R(\cos(2\pi t) + \omega \sin(2\pi t))$ and $t \in [0, 1]$ the winding number of the highest order term of the polynomial is $\mathcal{R}(a_n(q - q_0)^n, q_0) = n$. The intent is to prove that the entire polynomial has a nonzero winding number and then use Theorem 4.3. Therefore, over C_{q_0} , we have

$$|F(q) - a_n(q - q_0)^n|_{C_{q_0}} = \left| \sum_{\ell=0}^{n-1} a_{n-\ell}(q - q_0)^\ell \right|_{C_{q_0}} \quad (31)$$

$$\leq \sum_{\ell=0}^{n-1} |a_{n-\ell}| \left| (q - q_0)^\ell \right|_{C_{q_0}} \quad (32)$$

$$< R^n = a^n |a_n(q - q_0)^n|_{C_{q_0}} \quad (33)$$

Using Corollary 4.3, we obtain $\mathcal{R}(a_n(q - q_0)^n, q_0) = \mathcal{R}(F(C_{q_0}), q_0) = n$. From Theorem 4.3, the polynomial has a root. $\square$